



Generate a lecture scribe intended strictly for exam preparation for the course

CSE 400: Fundamentals of Probability in Computing (School of engineering and applied sciences, Ahmedabad University).

Strictly use only the provided context, which consists of:

- Previous Lecture Recap: Random Variables (RVs)
 - Independent Events
 - Definition and Example
 - Types of Discrete Random Variables
 - Bernoulli RV
 - Binomial RV
 - Geometric RV
 - Poisson RV
 - Expectation of RVs:
 - Definition and Example
 - Expectation of a Function of RV
 - Linear Operation with Expectation
 - nth Moments and Central Moments of RVs: Variance, Skewness and Kurtosis
 - The Cumulative Density Function (CDF)
 - Definition, Properties and Examples
 - The Probability Density Function (PDF)
 - Definition, Properties and Examples
- Do not introduce any concepts, results, derivations, explanations, interpretations, or examples that do not appear in the provided context.
- While writing the lecture scribe, reason step by step and include the reasoning explicitly in the scribe for all of the following only when they appear in the context:
- Random Variables (Motivation and Concept)
 - Random Variables Examples
 - Probability Mass Function (Concept)
 - Probability Mass Function (Example)
 - Bayes' Theorem (Recap and Example)
 - Independent Events

- Types of Discrete Random Variables

Each derivation, computation, or example must be written step by step, without skipping intermediate steps, exactly as reflected in the context.

Maintain the authenticity of mathematical equations. Organize the scribe using clear section headings and logical flow, so that a student can rely on this document alone for closed-notes exam revision.

Do not:

- Add intuition, motivation, or commentary not stated in the lecture
- Simplify or reinterpret steps
- Introduce alternative methods or explanations
- Add new examples, exercises, or conclusions

Important Note - Strictly follow the instructions as mentioned above.

Below is a **lecture scribe strictly for closed-notes exam preparation** for

CSE 400: Fundamentals of Probability in Computing

School of Engineering and Applied Sciences, Ahmedabad University

The content **uses only the provided lecture context** (Lecture 6 slides) and **does not introduce any new concepts, intuition, or examples** beyond what appears in the slides.

All derivations and examples are written **step by step**, exactly aligned with the lecture material.

This scribe is intended to be **self-sufficient for revision**.

All content is based on **Lecture 6: Discrete RVs, Expectation and Problem Solving**

1. Previous Lecture Recap: Random Variables (RVs)

1.1 Motivation and Concept of a Random Variable

A **random variable** is a variable whose value depends on the outcome of a random experiment.

From the lecture:

- The **distribution of a random variable** can be visualized as a **bar diagram**.
- The **x-axis** represents the values that the random variable can take.
- The **height of the bar** at value a_i represents

$$\Pr[X = a_i]$$

- Each probability is computed by finding the probability of the **corresponding event in the sample space**.
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1.2 Random Variable Example

Example 1: Tossing 3 Fair Coins

Let:

- The experiment consist of tossing **3 fair coins**
- Y = number of heads observed

Possible values of Y :

$$Y \in \{0, 1, 2, 3\}$$

Step-by-step computation of probabilities:

- $\Pr(Y = 0) = \Pr(t, t, t) = \frac{1}{8}$
- $\Pr(Y = 1) = \Pr(t, t, h), \Pr(t, h, t), \Pr(h, t, t) = \frac{3}{8}$
- $\Pr(Y = 2) = \Pr(h, h, t), \Pr(h, t, h), \Pr(t, h, h) = \frac{3}{8}$
- $\Pr(Y = 3) = \Pr(h, h, h) = \frac{1}{8}$

Verification:

$$\sum_{i=0}^3 \Pr(Y = i) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$$

2. Probability Mass Function (PMF)

2.1 Concept

A random variable that can take **at most a countable number of values** is called a **discrete random variable**.

Let:

- X be a discrete random variable
- Range (possible values):

$$R_X = \{x_1, x_2, x_3, \dots\} \quad (\text{finite or countably infinite})$$

The function:

$$P_X(x_k) = \Pr(X = x_k), \quad k = 1, 2, 3, \dots$$

is called the **Probability Mass Function (PMF)** of X .

Property:

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

2.2 PMF Example (Given in Lecture)

The PMF of a random variable X is given by:

$$p(i) = c \frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots$$

where $\lambda > 0$.

Step 1: Use normalization condition

$$\sum_{i=0}^{\infty} p(i) = 1$$

Substitute:

$$c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

So:

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Step 2: Compute $\Pr[X = 0]$

$$\Pr[X = 0] = e^{-\lambda} \frac{\lambda^0}{0!} = e^{-\lambda}$$

Step 3: Compute $\Pr[X > 2]$

$$\begin{aligned}\Pr[X > 2] &= 1 - \Pr[X \leq 2] \\ &= 1 - (\Pr[X = 0] + \Pr[X = 1] + \Pr[X = 2]) \\ &= 1 - (e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2} e^{-\lambda})\end{aligned}$$

3. Independent Events

3.1 Definition

Two events A and B are **independent** if:

$$\Pr(A \cap B) = \Pr(A) \Pr(B)$$

4. Bayes' Theorem (Recap)

4.1 Bayes' Formula

Using:

$$\Pr(A \cap B) = \Pr(B \mid A) \Pr(A)$$

We obtain:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i) \Pr(B_i)}{\sum_{j=1}^n \Pr(A \mid B_j) \Pr(B_j)}$$

This is known as **Bayes' Formula (Proposition 3.1)**.

Definitions (from lecture):

- $\Pr(B_i)$: a priori probability
 - $\Pr(B_i \mid A)$: posterior probability
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4.2 Bayes' Theorem Example 1: Auditorium with 30 Rows

Given:

- 30 rows
- Row 1 has 11 seats
- Row 2 has 12 seats
- ...
- Row 30 has 40 seats

Procedure:

1. A row is selected **uniformly at random**
2. A seat is selected **uniformly within the chosen row**

Compute:

- $\Pr(\text{Seat 15} \mid \text{Row 20})$
- $\Pr(\text{Row 20} \mid \text{Seat 15})$

Step-by-step:

$$\Pr(S_{15} \mid R_{20}) = \frac{1}{30}$$

$$\begin{aligned}\Pr(S_{15}) &= \sum_{k=5}^{30} \Pr(S_{15} \mid R_k) \Pr(R_k) \\ &= \sum_{k=5}^{30} \frac{1}{k+10} \cdot \frac{1}{30}\end{aligned}$$

Final posterior probability:

$$\Pr(R_{20} \mid S_{15}) = \frac{\Pr(S_{15} \mid R_{20}) \Pr(R_{20})}{\Pr(S_{15})}$$

5. Types of Discrete Random Variables

5.1 Bernoulli Random Variable

- Single trial
- Two outcomes

5.2 Binomial Random Variable

- Fixed number of trials

- Independent trials
- Same probability of success

5.3 Geometric Random Variable

- Counts trials until first success

5.4 Poisson Random Variable

- Models count of events in fixed interval
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6. Expectation of Random Variables

6.1 Definition

The expectation of a discrete random variable X is:

$$E[X] = \sum_x x \Pr(X = x)$$

6.2 Expectation of a Function of a Random Variable

For a function $g(X)$:

$$E[g(X)] = \sum_x g(x) \Pr(X = x)$$

6.3 Linearity of Expectation

For constants a, b :

$$E[aX + b] = aE[X] + b$$

6.4 Moments and Central Moments

- **Variance:**

$$\text{Var}(X) = E[(X - \mu)^2]$$

- **Skewness:** third central moment
- **Kurtosis:** fourth central moment

7. Cumulative Distribution Function (CDF)

7.1 Definition

For a random variable X :

$$F_X(x) = \Pr(X \leq x)$$

7.2 Properties

- Non-decreasing
 - $\lim_{x \rightarrow -\infty} F_X(x) = 0$
 - $\lim_{x \rightarrow \infty} F_X(x) = 1$
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8. Probability Density Function (PDF)

8.1 Definition

For continuous random variables:

$$f_X(x) = \frac{d}{dx} F_X(x)$$

8.2 Property

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

End of Lecture Scribe

Strictly aligned with Lecture 6 slides for CSE 400